Sammed Patil

Senior Backend Engineer | Python / Django / FastAPI

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Certifications

- AWS Certified Solutions Architect - Associate (SAA-C03)
- AWS Certified Cloud Practitioner (CLF-C02)

Professional Summary

- Backend Engineer with 4+ years building production applications for healthcare, sports and business clients using Python, Django and FastAPI. AWS certified, and hands-on from database design through to deployment.

Technical Skills

- Languages: Python, JavaScript, SQL
- Backend: Django 5.x, Django REST Framework, FastAPI, Pydantic, Celery, Django Channels (WebSockets)
- Databases: PostgreSQL, MongoDB, Redis, SQLite, ChromaDB
- Security: JWT with refresh rotation and blacklisting, RBAC, DRF throttling, HMAC-SHA256 webhook verification, file upload validation, OpenAPI
- AI/GenAI: OpenAI APIs, LangChain, RAG, ChromaDB vector search, HuggingFace embeddings, Ollama, prompt engineering, LLM provider routing and failover
- DevOps: AWS (EC2, S3, IAM, Lambda), Docker, Docker Compose, Nginx, Gunicorn, Daphne, GitHub Actions, CI/CD, Linux, DigitalOcean, Vercel
- Concepts: RESTful architecture, microservices, database design, query optimization, system design, MVC architecture

Professional Experience

AimedAt Solutions Pvt. Ltd. - Associate Software Engineer | Bengaluru

03/2022 - 10/2026

- Built 89 backend endpoints in Django REST Framework and FastAPI for separate patient, doctor and admin portals, all sharing one 39-model database schema instead of duplicating models per portal.
- Took a codebase with no automated tests to 78 passing tests, writing them test-first so every bug fix stays permanently protected against regression.
- Fixed a bug where a fighter's record showed 6 wins and 6 losses instead of 2 and 3, because two database joins multiplied each other; rebuilt the query with correlated subqueries and added 9 tests.
- Built a background job system on Celery and Redis that automatically sends patients diet, exercise and medication reminders on 12-hour and weekly schedules, retries failed push deliveries 3 times, and removes invalid device tokens.
- Fixed a client complaint about notification overload with a Redis policy limiting each doctor to 25 alerts per day and blocking duplicates within 24 hours, applied centrally to every notification type except chat.
- Built a reusable audit base model applied across 37 tables, automatically recording who created and last changed every record without threading user context through every function.
- Secured the API with role-based access control for 4 roles (superadmin, admin, doctor, patient), login tokens that refresh automatically and can be revoked on logout, and rate limits that block OTP abuse per phone number.
- Hardened transport security with HSTS, frame denial, content-type sniffing protection and secure cookie flags, with proxy trust set so client IPs resolve correctly behind Nginx.
- Fixed a payment security flaw where a user could request a Rs.1 charge for a Rs.499 registration and have it accepted as paid; the server now sets all prices and converts rupees to paise exactly.
- Added payment state rules so a refunded or already-paid order cannot be silently reverted by a repeated confirmation request, and made payment and application updates save together so a crash cannot leave them disagreeing.

- Containerised the backend into 6 services (Nginx, Gunicorn, Daphne, Celery worker and scheduler, Redis) with health checks and HTTPS, and set up CI/CD that builds and deploys to a DigitalOcean server.
- Found 12 of 20 database migration files were excluded from version control, so any developer cloning the repo got an empty database; restored them and confirmed a clean clone now builds the full 49-table schema.

Key Projects

RoninFC - Combat sports and athlete management platform | Django, DRF, PostgreSQL 16, Razorpay, Celery, Redis

- Sole backend owner for a 16-module platform serving 147 endpoints across 53 database models, storing all money in exact decimal fields and blocking deletion of records with financial or legal significance.
- Validated file uploads by actual content rather than file extension, using image and PDF allow-lists with real decoding, so a renamed executable cannot be uploaded.
- Integrated Razorpay for registration payments through 4 endpoints (create, verify, history, webhook), identifying each payment by the gateway's order ID and restricting lookups to the logged-in user to prevent IDOR.
- Chained database lookups across three levels of related records (match, event, fighter profile), collapsing what would otherwise be dozens of separate queries per page load.

SamAgent OS - AI platform for a business services firm | FastAPI, MongoDB, React, AWS S3, Vercel, Docker

- Built admin-configurable AI model routing so the LLM provider and model can be changed for all 13 AI features from a database setting, with no redeploy, plus failover to any OpenAI-compatible provider.
- Designed AI failover that retries on rate limits, server errors and timeouts but fails fast on authentication errors, so a misconfigured API key reports a clear error instead of being hidden by a fallback model.
- Stored third-party API credentials encrypted at rest and returned masked values in every response, so a dashboard screenshot cannot leak a live API key.

MHealth - Chronic disease management platform | Django, DRF, Celery, Redis, Channels, Firebase Cloud Messaging

- Delivered the doctor, patient and admin APIs on one shared 39-model schema rather than duplicating models per portal, with auto-generated OpenAPI docs and 110 automated tests covering the highest-risk logic.
- Built real-time messaging: Firebase Cloud Messaging reaches all of a patient's devices in one call and removes dead device tokens automatically, while JWT-authenticated WebSockets over Django Channels stay open behind Nginx.
- Built a translation engine for a Kannada-first healthcare platform that invalidates only the specific fields that were edited, instead of flagging an entire record as stale on any change.

RAG Document Assistant | FastAPI, LangChain, ChromaDB, HuggingFace embeddings, Ollama

- Built a document question-answering assistant that finds relevant passages before answering: PDFs are split into chunks, stored as vectors in ChromaDB, and every answer cites the page and filename it came from.

Education

- Master of Computer Applications (MCA) - Sambhram Academy of Management Studies, Bangalore City University, 10/2021
- Bachelor of Computer Applications (BCA) - SJPN Trust's BCA College, RCU University, Belagavi, 05/2019